Santhosh S

+91-9597363959 | santhosh.sampath2024@vitstudent.ac.in | LinkedIn | GitHub

OBJECTIVE

Second-year Electronics & Computer Engineering undergraduate at VIT Chennai (CGPA: 8.67/10), with hands-on research experience in deep learning and computer vision. Proficient in building end-to-end segmentation and detection pipelines in PyTorch. Seeking to strengthen ML fundamentals — probability, linear algebra, and optimization — under Amazon scientists' mentorship at MLSS 2026.

EDUCATION

B.Tech, Electronics & Computer Engineering — Vellore Institute of Technology, Chennai | CGPA: 8.67 / 10 — 2024 – 2028

Intermediate / +2 — Narayana E-techno School, Poonamallee | 78% — 2022 – 2024

CBSE / 10th — Lalaji Memorial Omega International School, Chennai | 94% — 2021 – 2022

TECHNICAL SKILLS

Programming: Python, Java, C, C++

ML Libraries & Tools: PyTorch, NumPy, Pandas, Scikit-learn, OpenCV, Albumentations, Matplotlib

ML Concepts: CNNs, UNet++, Object Detection, Image Segmentation, Transfer Learning, TTA, Data Augmentation, Model Evaluation (IoU, Dice, F1, Cohen's Kappa)

Geospatial Tools: QGIS — NDWI calculation, spatial analysis, satellite data processing

INTERNSHIP

ESG Sentinel Pvt. Ltd. — Full Stack Developer Intern — May 2026 – June 2026

- Contributed to the end-to-end development of a live enterprise compliance platform, working across frontend, backend, and database layers
- Built and integrated new features on a production codebase as part of an active development team
- Adopted professional Git & GitHub workflows — branching, commits, pull requests — in a real collaborative engineering environment

PROJECTS

Glacial Lake Detection using Deep Learning | GLOF Challenge – SNU Chennai

- Built a **UNet++ segmentation pipeline with EfficientNet-B3 encoder** for detecting glacial lakes from multispectral satellite imagery (RGB + NDWI as 4th input channel)
- Implemented combined Focal + Dice + Boundary loss to handle extreme class imbalance in sparse lake masks; applied CutMix augmentation and multi-checkpoint TTA ensemble for robust inference
- Evaluated using IoU, Dice, Precision, Recall, F1-Score, Specificity, and Cohen's Kappa; applied morphological post-processing to reduce false positives

YOLOv8n with Efficient Channel Attention (ECA) for Floor Plan Analysis | Computer Vision Research

- Developed **YOLOv8n augmented with ECA attention mechanism** (Wang et al., CVPR 2020) for structural element detection in architectural floor plans; integrated attention with minimal parameter overhead
- Evaluated on SESYD and CVC-FP benchmark datasets using mAP, IoU, Precision, and Recall; documented architecture and results in a research paper submission

Water Body Mapping using NDWI | Remote Sensing & Geospatial Research

- Mapped and quantified water bodies from multispectral satellite imagery by computing NDWI in QGIS; applied spatial analysis for water resource management
- Completed as the core project deliverable during the Remote Sensing & GIS Internship at India Space Academy (Jul – Aug 2025)

CERTIFICATIONS

Remote Sensing & GIS Internship Certificate — India Space Academy — Jul – Aug 2025

4-week project on water body detection using NDWI from multispectral satellite imagery for water resource management applications.

Oracle Cloud Infrastructure 2025 Certified Generative AI Professional — Oracle University — Jul 2025

Validates proficiency in Generative AI concepts, Oracle Cloud Infrastructure (OCI) services, and AI-driven application development.